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Introduction

In this assignment we will go through the task of designing a database backup, recovery and turnover management plan.

Backup Models – Robust

Robust Model	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
2 AM	Full	Full	Full	Full	Full	Full	Full
3 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
4 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
5 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
7 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
8 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
10 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
11 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
12 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
1 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
2 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
4 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
5 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
6 PM	Full	Full	Full	Full	Full	Full	Full
7 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
8 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
10 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
11 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction

This model is extremely extensive with multiple full log backups and multiple transaction log backups to ensure proper server runtime.

Backup Model – Intermediate

Robust Model	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12 AM							
1 AM							
2 AM	Full	Full	Full	Full	Full	Full	Full
3 AM							
4 AM							
5 AM							
6 AM							
7 AM							
8 AM	Full	Full	Full	Full	Full	Full	Full
9 AM							
10 AM							
11 AM							
12 PM							
1 PM							
2 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3 PM							
4 PM							
5 PM							
6 PM							
7 PM							
8 PM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
9 PM							
10 PM							
11 PM							

This model is basic with only two transaction log backups a day and two full backups per day.

Backup Model – Simple

Robust Model	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12 AM							
1 AM							
2 AM	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction	Transaction
3 AM							
4 AM							
5 AM							
6 AM							
7 AM							
8 AM							
9 AM							
10 AM							
11 AM							
12 PM							
1 PM							
2 PM	Full						
3 PM							
4 PM							
5 PM							
6 PM							
7 PM							
8 PM							
9 PM							
10 PM							
11 PM							

The simple model is what a Dba would use as it requires a full back up only once a week and only one transaction log backup per day.

Metrics

1. Backup Duration-Time required to complete full and log backups
2. Restoration Time- Time from failure detection to full recovery
3. Data Size- Total volume of data in the system
4. Downtime Impact- Number of users affected and duration
5. Business Cost Estimate- Revenue or productivity loss due to outage
6. Failure Frequency- Average number of incidents per quarter
7. Recovery Start Time-Timestamp of recovery initiation

Turnover/Change Controls

Turnover management refers to the structured process of introducing changes into a production database environment. Effective turnover management ensures that these updates occur with minimal disruption to users and systems, while maintaining a clear audit trail of approvals, responsibilities, and risk assessments.

There are two primary categories of turnover, routine (or simple) turnovers and complex (or high-risk) turnovers.

Routine turnovers are typically scheduled for late afternoon hours, such as 5 PM, when system usage is lower and any minor disruptions are unlikely to impact business operations significantly. Simple turnover usually does not require a database backup beforehand, as the changes are considered low risk and easily reversible.

Complex turnover on the other hand involves significant changes that, if unsuccessful, could negatively affect system stability or data integrity. These are scheduled during low-traffic periods typically at midnight to reduce the likelihood of user disruption. The change request form for a complex turnover must be submitted and approved at least one day in advance. Unlike simple turnover, complex turnovers require a full database backup prior to implementation to ensure a secure rollback option is available. The DBA must be consulted to evaluate whether a secondary backup should be taken after the change is complete. During the actual turnover, both the requestor and technical staff must be present to monitor the process and respond to any unexpected issues in real time.

In both cases, the business user's approval is mandatory. This ensures that those responsible for the day-to-day operation of the system are aware of and agree to the change. This shared accountability fosters transparency, preparedness, and operational integrity. By enforcing these structured turnover procedures, organizations reduce risk and enhance the reliability of their database environments.

Change Request Form

Section 1 – Submitter Details

Field	Input
Change Request ID (CR#)	CR-2025-01
Type of Request	☐ Enhancement ☐ Bug Fix ☐ Infrastructure ☐ Other: _____
Project or System Name	_____
Submitted By	_____ Date: ____ / ____ / _____
Title of Change	_____
Detailed Description	_____
Required Completion Date	____ / ____ / _____
Priority Level	☐ Low ☐ Moderate ☐ High ☐ Urgent
Purpose of Request	
Affected Modules or Items	_____
Additional Comments	
Attachments Provided?	☐ Yes ☐ No Link: _____

Section 2 – Initial review by project manager

Field	Input
Estimated Work Hours	_____ Hours
Estimated Duration	_____ Days
Schedule Impact	_____
Budget or Cost Impact	_____
Resources Required	_____
PM Notes & Assessment	

Section 3 – Change control board decision

Field	Input
Final Decision	☐ Approved ☐ Approved w/ Conditions ☐ Rejected ☐ Need More Info
Review Date	___ / ___ / _____
Justification / Conditions	
Additional Notes	_____